

Indian Institute of Technology (Indian School of Mines) Dhanbad

Winter End Semester Question Paper

Examination: B.Tech (1st Year, Sections A, B, C, & D)

Time: 2 Hours

Discipline: Mechanical Engineering

Full Marks: 50

Subject: Manufacturing Process (NMES 101)

Session: 2024 - 25 (Winter)

(Answer all the questions. Provide the answer to each part in one continuous space without break)

Part – I (15 marks)

1. In an arc welding process, the voltage and current are 30 V and 200 A, respectively. The cross-sectional area of the joint is 20 mm² and the welding speed is 5 mm/s. The heat required to melt the material is 20 J/mm³. Determine the percentage of heat lost to the surrounding during the welding process rounded off to 2 decimal places. [3]
2. In a direct current arc welding process, the power source has an open circuit voltage of 100 V and short circuit current of 1000 A. Assume a linear relationship between voltage and current. The arc voltage (V) varies with the arc length (L) as $V = 10 + 5L$ where V is in volts and L is in mm. Determine the maximum available arc power during the process in kVA (in integer). [4]
3. Two mild steel plates of similar thickness, in butt-joint configuration, are welded by gas tungsten arc welding process using the following welding parameters: Welding voltage = 20 V, Welding current = 150 A, and Welding speed = 5 mm/s. A filler wire of the same mild steel material having 3 mm diameter is used in this welding process. The filler wire feed rate is selected such that the final weld bead is composed of 60% volume of filler and 40% volume of plate material. The heat required to melt the mild steel material is 10 J/mm³. The heat transfer factor is 0.70 and melting factor is 0.60. Determine the feed rate of the filler wire in mm/s (round off to one decimal place). [4]
4. The voltage - length characteristic of a direct current arc in an arc welding process is $V = (100 + 40L)$, where L is the length of the arc in mm and V is arc voltage in volts. During a welding operation, the arc length varies between 1.0 and 2.0 mm and the welding current is the range 200 – 250 A. Assuming a linear power source, determine the short circuit current. [4]

Part – II (25 marks)

5. Write three input parameters and two output parameters of turning in the lathe machine. [2]
6. Explain servomotor and stepper motor. [3]
7. Write five numbers of machine tool drives. [2]
8. Write 10 parameters of specification of lathe machine. [3]
9. In an orthogonal machining process, the following data are observed: Uncut chip thickness = 0.127 mm; Width of cut = 6.35 mm; Cutting speed = 2 m/s; Rake angle = 10°; Cutting force = 567 N; Thrust force = 227 N; Chip thickness = 0.228 mm. Determine: (i) Shear angle, (ii) the friction angle, (iii) shear stress (in N/mm²) along the shear plane and (iv) the power (in kW) for cutting operation. [6]

10. The following equation for tool life is given for a turning operation:

$$VT^{0.13}f^{0.77}d^{0.37} = C$$

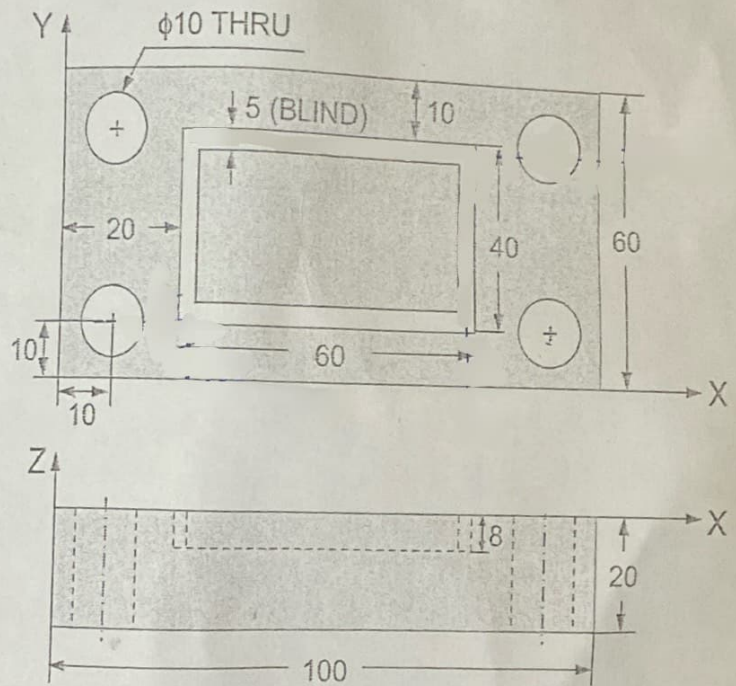
[3]

Tool life of 60 minute was obtained while cutting at $V = 30$ m/min, $f = 0.3$ mm/rev and $d = 2.5$ mm. Determine the change in tool life if the cutting speed, feed, and depth of cut are increased by 20% each.

11. In the year 2050, aboard a space-based manufacturing station, an autonomous ECM unit is assigned to cut a circular hole through a 12 mm thick aluminum plate for a deep-space rover component. The hole has a diameter of 20 mm, and the operation will use a current of 1200 A with a system efficiency of 95%. Determine the feed rate and the time required to complete the cut. $C = 3.44 \times 10^{-2} \text{ mm}^2/\text{A-sec}$ [3]
12. A defence research laboratory is developing an automated abrasive jet system to remove unwanted material from a titanium alloy surface used in aerospace applications. The system ejects abrasive particles at a velocity of 250 m/s, with a mass flow rate of 1.2 kg/min. The abrasives are made of a dense ceramic material with a density of 4.5 kg/mm³. The titanium alloy surface has a hardness of 3000 N/mm². You being an engineer need to determine the material removal rate (MRR) to ensure that the process is efficient yet does not compromise the structural integrity of the component. [3]

Part – III (10 marks)

13. Write a G&M code program for cutting a rectangular blind slot and four symmetric through holes on a 100×60×20 mm³ rectangular plate, as shown in the adjacent figure. Available end-milling tool (T02) has diameter of 5 mm and drill (T06) has diameter of 10 mm. [8]



14. In robotic manufacturing, explain what is “cobot” and how it ensures safety of the human operators. [2]